

Dear David

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NSW 2747.

Please find enclosed 4 separate sets of information

- (1) These are the instructions I supply with my software describing the hardware mod.
Please note that the MAX 232 chip is external in a box. (errors corrected)
- (2) Issue 29 of HUV2UG magazine describing Joe Leon's modification which puts the MAX 232 chip on a board inside the disk controller.
- (3) Issue 30 of HUV2UG magazine correcting errors in (1)/(2) and giving more details of Joe's version.
- (4) A complete schematic of Joe Leon's version - no errors

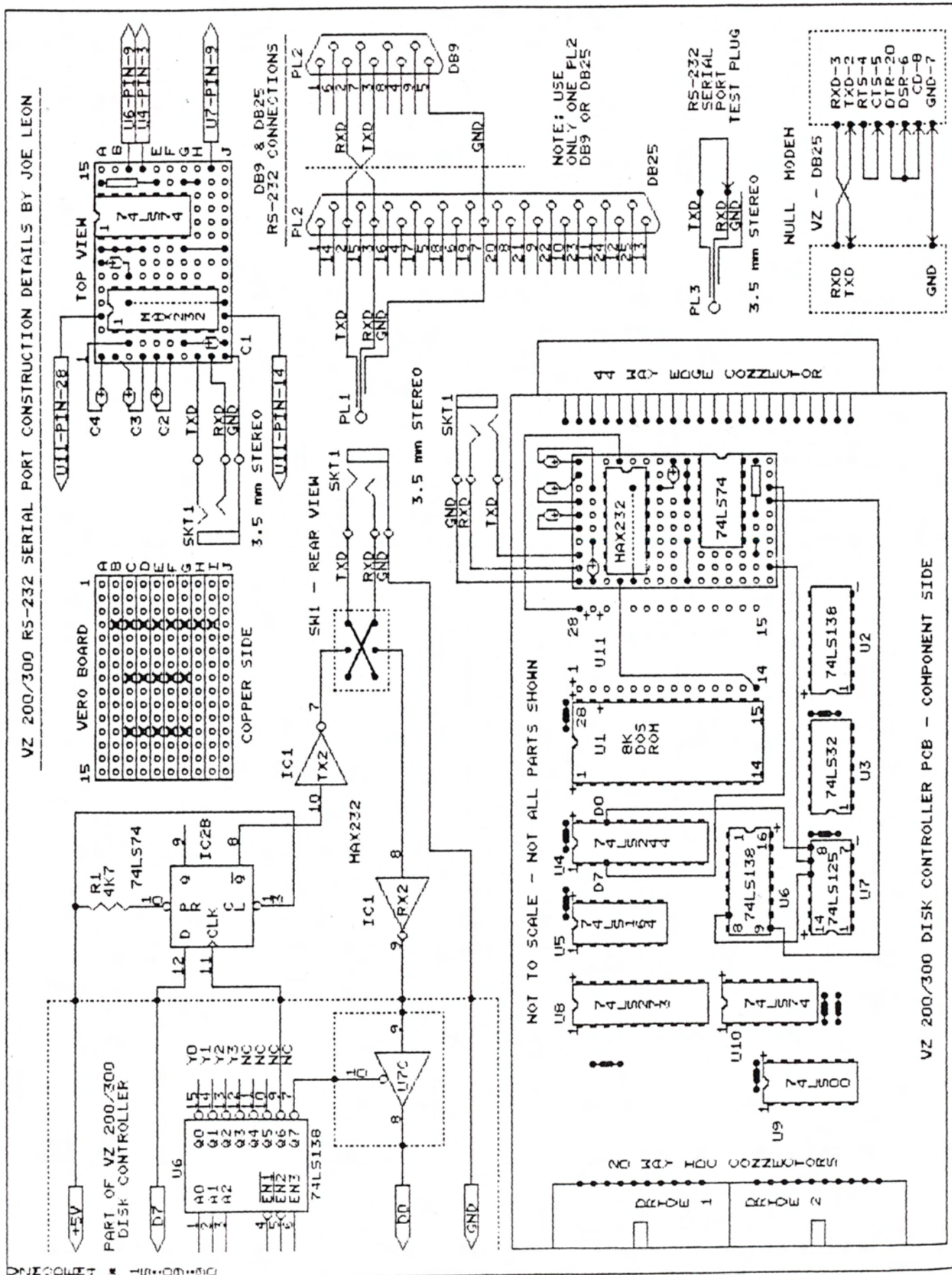
To test the mod you will need the software to run it. Simply wire the TXD to the RXD and what you send will be printed on your screen.

You can send/receive any file & save them to disk.

Feel free to ring me Sat 12-6pm for any info.
or to try out your device. 047-304880.

Yours - computers

Pete Hickman



HARDWARE MODIFICATION

This program controls the hardware. The program will not function correctly without the hardware. The hardware modification is useless without the software. The modification is performed on the disk controller using two spare output ports of the existing decoder (74LS138). Two new ICs are piggy backed onto existing ICs inside the disk controller. One IC is housed in a separate box. Joe Leon has re-designed this so that they all fit inside the disk controller. See issues 29 & 30 of HVVZUG magazine.

PARTS REQUIRED

- 1 - 74LS74 (U3A)
- 1 - 74LS125 (U7A)
- 1 - MAX232
- 2 - 4.7K 1/4w resistors
- 2 - 10uF 6.3V capacitors
- 3 - 10uF 16V capacitors
- 1 - small box 50mmx90mm

CONSTRUCTION DETAILS

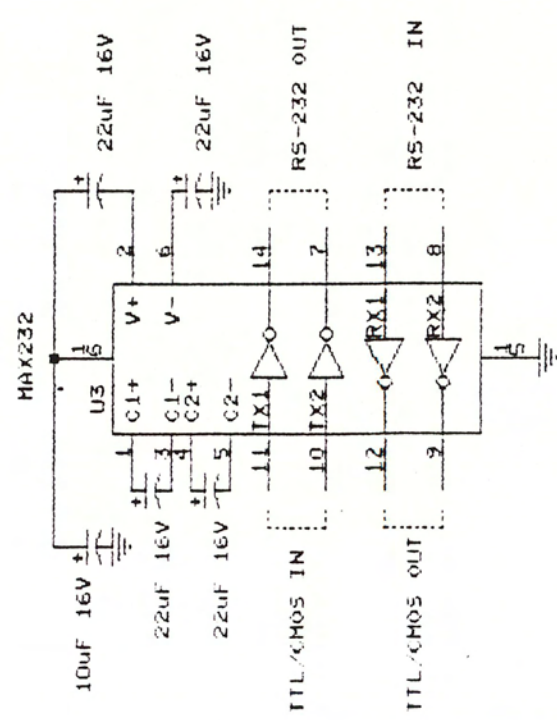
Remove cover and metal shield from disk controller. Solder one wire onto pin 7 & one wire onto pin 9 of U6 (74LS138). Leave the wire ends loose for later connection. Bend the pins of U3A and U7A at right angles, but don't touch pins 7 & 14. Mount the new ICs. Solder pin 7 & pin 14 of U7A (74LS125) onto the equivalent pins of U7. Repeat this step for pins U3A & U3. Connect the wire from pin 7 of U6 to pin 10 of U7A. Connect the wire from pin 9 of U6 to pin 11 of U3A. Connect pin 8 of U7A to pin 13 of U4 (74LS244). Connect pin 12 of U3A to pin 3 of U4. Connect pin 13 of U3A to pin 14 of U3A. Connect one 4.7k resistor to pin 10 of U3A and the other resistor to pin 11 of U3A. Connect the loose ends of the resistors together and to 5 volts on pin 16 of U6 (74LS138). The MAX232 IC is mounted on a small piece of vero board with 5 capacitors and can be put into a small box. The reason that this IC is external to the disk controller is because the 5 capacitors associated with it take up too much room to easily fit into the controller.

MAX232 CONNECTIONS

- Connect a 10uF 6.3V capacitor between pin 1 and pin 3 - +ve to pin 1.
1. Connect a 10uF 16V capacitor between pin 2 and pin 16 - +ve to pin 16.
 2. Connect a 10uF 6.3V capacitor between pin 16 and ground - +ve to pin 16.
 3. Connect a 10uF 16V capacitor between pin 4 and pin 5 - +ve to pin 4.
 4. Connect a 10uF 16V capacitor between pin 6 and ground - +ve to ground.

CONNECTIONS BETWEEN DISK CONTROLLER AND MAX232

Four wires are used to connect to the MAX232 via a small hole cut in the back of the controller near the drive connections. These wires need to be secured in some way as a strain relief. Look at printer interface for idea. Connect between pin 9 of the MAX232 and pin 9 of U7A. Connect between pin 10 of the MAX232 and pin 8 of U3A. Two connections are required for +5 volts and ground. When replacing the metal shield the pins that you bent out on the ICs can be bent back down, so that they don't interfere with the shield. Make sure that these pins do not touch the lower IC on which they have been piggy-backed.



VZ SERIAL INTERFACE CONTINUED 29/16

LIKE MANY DSE VZ TERMINAL USERS PETER WAS FRUSTRATED BY ITS SEVERE LIMITATIONS SO IN CORROBORATION WITH HIS BROTHER ANDY THEY HAVE REDESIGNED BOTH THE SOFTWARE AND HARDWARE. AS DSE TERMINALS ARE PRACTICALLY IMPOSSIBLE TO BUY THIS PROJECT REPRESENTS A MUCH CHEAPER AND MORE FUNCTIONAL ALTERNATIVE.

NOTE : THE SOFTWARE FOR THIS SERIAL INTERFACE IS DISK BASED ONLY.

THE PARTS COUNT IS QUITE SMALL, JUST THREE CHIPS, 5 CAPACITORS AND ONE OR TWO CONNECTOR PLUGS. IT IS POSSIBLE TO BUILD THE CIRCUIT INSIDE THE DISK CONTROLLER BY PIGGYBACKING THE 74LS74 AND 74LS125 ON EXISTING CHIPS AND PLACING MAX232 CHIP NEAR THE EDGE CONNECTOR.

A 3.5MM STEREO SOCKET COULD BE USED AS ONLY THREE WIRES ARE INVOLVED, TXD (TRANSMIT DATA), RXD (RECEIVE DATA) AND GROUND. IT WOULD MAKE FOR A NICE COMPACT UNIT WITH NO BOXES TO PLUG IN. ALTERNATIVELY A SMALL BOX COULD BE USED BUT WOULD REQUIRE SIX WIRES, +5V, GND, Q6, Q7 (U6-74LS138), DATA LINE 0 & 7. IT CAN ALSO BE BUILT INTO VZ 200/300 BUT WOULD REQUIRE DUPLICATING DECODING CIRCUITRY.

THE 74LS138 (U6) DECODER IN THE DISKCONTROLLER USES ONLY FIRST FOUR OUTPUTS SO FOUR ARE SPARE AND THE LAST TWO OUTPUTS ARE USED FOR CONTROLLING THE SERIAL INTERFACE WHICH SAVES TWO EXTRA CHIPS. THE 74LS74 'D' TYPE FLIP FLOP PASSES DATA TO A SECTION OF A MAX232 (U3-T2) CHIP FOR TRANSMITTING TO A MODEM OR ANOTHER COMPUTER.

FOR RECEIVING A SECTION OF MAX232 (U3-R2) IS USED WHICH PASSES DATA VIA A 74LS125 (U2C) TRISTATE BUFFER TO THE VZ. DATA LINE ZERO (0) IS USED FOR RECEIVING DATA WHILE DATA LINE SEVEN (7) IS USED FOR TRANSMITTING DATA.

THE MAX232 CHIP HAS TWO TRANSMITTERS AND TWO RECEIVERS AND ONLY ONE OF EACH IS NEEDED FOR THE SERIAL INTERFACE. ONLY ONE OF TWO FLIP FLOPS (74LS74) AND ONE OF FOUR TRISTATE BUFFERS (74LS125) ARE USED. THAT MEANS A SECOND SERIAL INTERFACE CAN BE BUILT FOR PRACTICALLY NO ADDITIONAL EXPENSE. NORMALLY A DB25 PIN OR DB9 PIN PLUG IS USED FOR CONNECTING TO A MODEM OR ANOTHER COMPUTER.

THE SECOND SERIAL INTERFACE IS SHOWN BELOW THE FIRST ONE AND SHARES COMMON SIGNALS LIKE +5V, GND, Q6 & Q7. D0, D7 AND GND WOULD COME FROM SECOND VZ VIA A 3.MM SOCKET WHICH IS SMALL AND EASY TO FIT. THE CONNECTION TO A MODEM OR ANOTHER COMPUTER WOULD NORMALLY BE VIA A DB25 OR DB9 CONNECTOR AND CONNECTIONS TO BOTH ARE SHOWN BUT ONLY ONE IS USED DEPENDING ON CONNECTOR YOU PLUG INTO.

THE MAX232 BLOCK DIAGRAM SHOWS THE FUNCTIONAL OPERATION OF THE CHIP. FOR COMPACTNESS TANTALUM CAPACITORS CAN BE USED EXSPECIALLY IF BUILDING UNIT INSIDE DISK CONTROLLER. A SMALL PIECE OF VEROBOARD CAN BE USED TO MOUNT THE MAX232 CHIP AND CAPACITORS AND OR OTHER TWO CHIPS AS WELL.

THE Q6 & Q7 SIGNALS ARE TAKEN DIRECT FROM THE 74LS138 DECODER IN THE DISK CONTROLLER. ALL UNUSED INPUTS ON 74LS74 AND 74LS125 MUST BE TIED HIGH OR UNWANTED OSCILLATION COULD OCCUR. THERE ARE THREE CONSTRUCTION OPTIONS. INSIDE DISK CONTROLLER, EXTERNAL BOX OR INSIDE THE VZ AND I LEAVE IT TO YOU ON WHICH OPTION YOU ADOPT.

NOTE : YOU'LL NEED ANOTHER COMPUTER OR VZ WITH A RS232 SERIAL INTERFACE TO TEST UNIT OUT.

29/17 VZ SERIAL INTERFACE CONTINUED

BEFORE PROCEEDING TO TEST UNIT THERE ARE SEVERAL THINGS THAT NEED EXPLAINING. WE'LL START WITH DB25 AND DB9. YOU'LL NOTE THAT ON DB25 TXD IS CONNECTED TO PIN 2 WHILE RXD IS CONNECTED TO PIN 3. THE REVERSE IS TRUE FOR DB9. SOME EQUIPMENT COULD COME WITH DB25 OR DB9 CONNECTORS AND MY IBM PC COMPATIBLE HAS TWO SERIAL PORTS HAVING BOTH CONNECTORS.

AS FAR AS THE VZ IS CONCERNED ONLY THREE SIGNALS ARE USED WHICH ARE TXD, RXD AND GND. CONNECTING TO A MODEM IS STRAIGHT-FORWARD. JUST USE A DB25 OR DB9 CONNECTOR AND WIRE UP THE THREE SIGNALS ACCORDING TO THE CONVENTION. ON PAGE 15 BOTTOM LEFT YOU'LL NOTE TWO DIAGRAMS ON HOW TO CONNECT VZ TO A MODEM OR ANOTHER VZ.

THE TWO TOP RIGHT DIAGRAMS DEPICT FIRST AGAIN CONNECTING COMPUTER TO A MODEM. THE NULL MODEM DIAGRAM IS HOW THE VZ AND ANOTHER COMPUTER LIKE THE IBM PC SHOULD BE CONNECTED. THE OTHER SIGNALS ARE SHOWN FOR CLARITY.

NOTE: THE MATERIAL FOR THIS PROJECT CAME FROM PETER AND ANDY HICKMAN WHO DID AN EXCELLENT JOB IN DESIGNING BOTH THE SOFTWARE AND HARDWARE. I WOULD LIKE TO THANK LOCAL MEMBERS FOR THEIR HELP AND RESEARCH WITH ABOVE PROJECT. ED.

FUNCTION KEYS . . .

IN THE SOFTWARE PETER HAS MADE PROVISION FOR EXTRA FUNCTION KEYS. THEY CAN BE USED WITH ABOVE SOFTWARE OR YOU COULD WRITE YOUR OWN WHICH CAN GIVE YOU UP TO 12 EXTRA FUNCTIONS, FOUR PER KEY LIKE SOME OF THE OTHER KEYS ON THE VZ. ALL THE DISK WORDS COULD BE ACTIVATED JUST BY A KEY PRESS OR TWO.

AS YOU'LL NOTE BY THE KEYBOARD MATRIX THERE ARE THREE PHYSICAL LOCATIONS EACH MARKED BY THREE ASTERICKS WHERE KEYS HAVE NOT BEEN CONNECTED. IT IS FAIRLY SIMPLE TO CONNECT THREE FUNCTION KEYS IN THOSE LOCATIONS. THE BOTTOM RIGHT DIAGRAM SHOWS WHERE THE CONNECTIONS ARE MADE.

I WOULD SUGGEST ROUND PUSHBUTTON KEYS WITH ABOUT 3/8" TOPS AS SOLD BY VARIOUS ELECTRONIC SHOPS. THE KEYS COULD BE MOUNTED ON A NARROW PIECE OF VEROBOARD AND IN THE CASE OF VZ300 PLACED JUST ABOVE THE KEYBOARD ON THE FLAT AND MOUNTED FROM UNDERNEATH. USING ROUND KEYS MAKES IT EASIER TO DRILL HOLES TO FIT THAN WOULD BE THE CASE FOR SQUARE TOPPED KEYS.

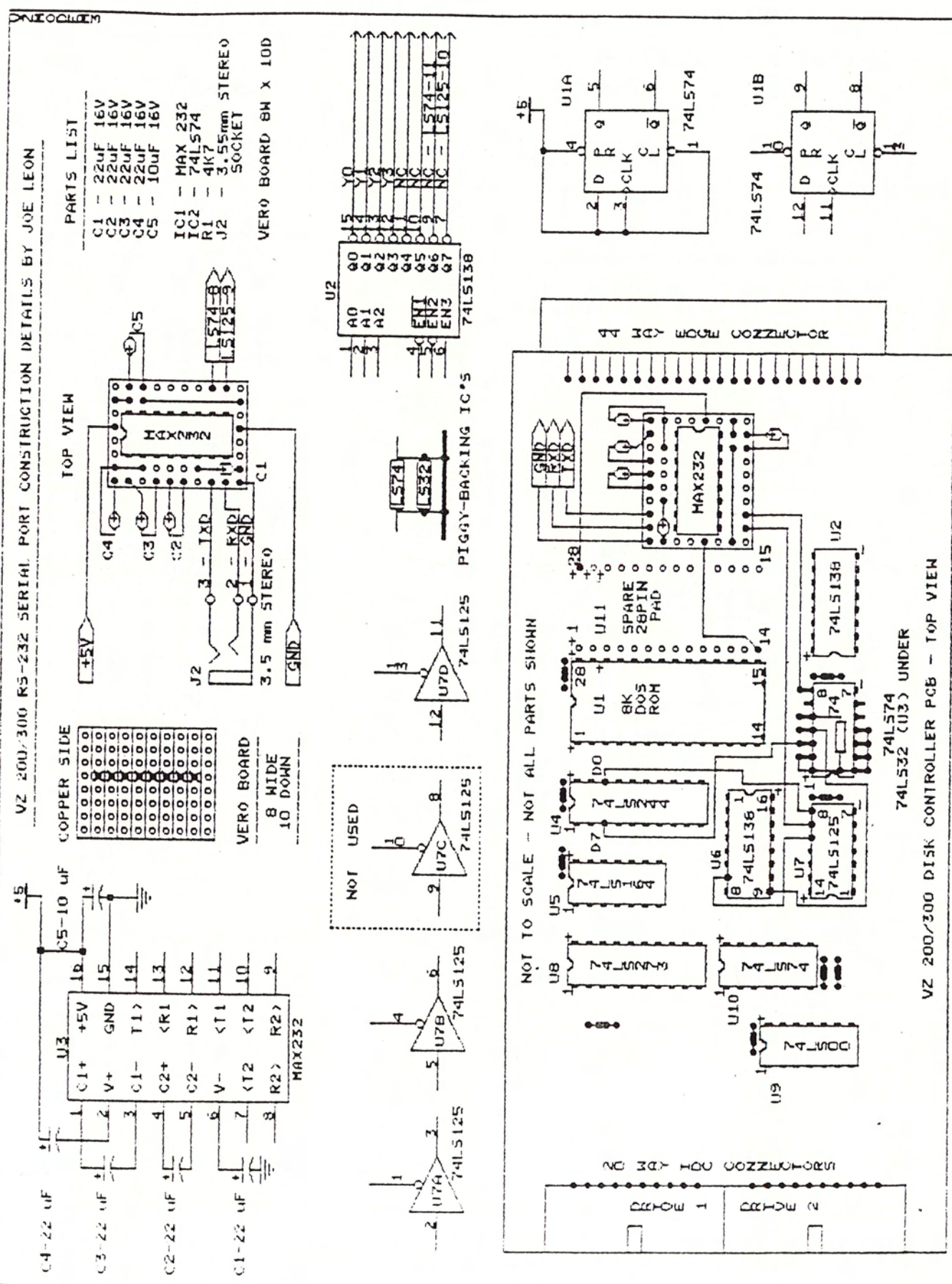
CONCLUSION . . .

WHEN COMPLETED, THIS VZ 200/300 RS-232 SERIAL INTERFACE WILL ALLOW YOU TO CONNECT IT TO A MODEM, ANOTHER COMPUTER AND TRANSMIT, RECEIVE AT 300 BAUD. DATA CAN BE PRELOADED FROM DISK BEFORE TRANSMITTING TO SAVE LOG ON TIME. THE SOFTWARE IS VERY VERSATILE AND PETER HAS GONE TO GREAT LENGTHS TO PROVIDE AS MANY FUNCTIONS AS POSSIBLE.

IT IS REFRESHING TO SEE NEW AND USEFUL PROJECTS MAKING AN APPEARANCE FOR THE VZ CONSIDERING THE PRESENT STATE OF THE VZ. WHILEVER THERE ARE PERSONS LIKE PETER AND ANDY HICKMAN AROUND THE VZ WILL SURVIVE. KEEP IT UP FELLAS.

FOR MORE INFORMATION ON ABOVE PROJECT OR SOFTWARE CONTACT PETER HICKMAN. FOR ADDRESS SEE HIS AD ON PAGE 18.

WZ 200/300 R5-232 SERIAL PORT CONSTRUCTION DETAILS BY JOE LEON



CORRECTIONS:

VZ JOURNAL # 29 - THE TXD AND RXD SIGNALS ON DB9 & DB25 CONNECTORS WERE TRANSPOSED ON PAGE 14 WHILE THOSE ON PAGE 15 WERE CORRECT.

2ND SERIAL INTERFACE ON PAGE 14 - ALL SIGNALS LEFT OF DASHED LINE INCLUDING Q6 & Q7 WITH THE EXCEPTION OF +5V MUST GO TO 2ND VZ'S DISK CONTROLLER.

THERE WERE ALSO SOME ERRORS WITH CONSTRUCTION DETAILS THAT CAME WITH THE SOFTWARE.

- 1) MAX232 CHIP - PIN 7 (TXD) GOES TO PIN 2 OF DB25 CONNECTOR.
PIN 8 (RXD) GOES TO PIN 3 OF DB25 CONNECTOR.
- 2) 74LS47 MENTIONED IN THE INSTRUCTIONS AS U3A SHOULD BE 74LS74 AND IS A NEW CHIP. DO NOT CONFUSE IT WITH U10-74LS74 IN THE DISK CONTROLLER.
- 3) U6-74LS138 - PIN 16 HAS +5V NOT PIN 14 WHICH IS OUTPUT Q1.

CONSTRUCTION DETAILS:

IF YOU ARE BUILDING ONLY SINGLE RS-232 PORT THEN IT CAN ALL FIT INSIDE THE DISK CONTROLLER QUITE EASILY AND ONE LESS IC IS REQUIRED, NAMELY THE 74LS125.

MAX232 PINOUT: IT SHOWS CAPACITOR CONNECTIONS AND THEIR ORIENTATION. BOTH ELECTROLYTIC AND TANTALUM CAPACITORS ARE SUITABLE WITH THE LATTER BEING SMALLER BUT MORE EXPENSIVE. THE CAPACITOR VOLTAGE RATINGS MIGHT NEED EXPLAINING TO SOME CONSTRUCTORS.

HIGHER VOLTAGE RATINGS CAN BE USED BUT NOT LOWER AND YOU MAY FIND IT DIFFICULT TO GET 16V TYPE AND UP TO 35V ARE OK, BUT THE SMALLER THE VOLTAGE RATING, THE SMALLER THE CAPACITOR NORMALLY AND SIZE IS IMPORTANT IN THIS PROJECT. OK TO USE 10UF-22UF FOR C1-C4 AS THE CAPACITY VALUE IS NOT CRITICAL.

VERO BOARD: ONLY A SMALL PIECE IS REQUIRED, 8 HOLES ACROSS AND TEN STRIPS DOWN. CUT 8 TRACKS ON THE VERO BOARD AS INDICATED BY THE X'S. FIT TWO LINK WIRES NEXT. SOLDER THE THREE WIRES FOR J2 FROM THE COPPER SIDE NEXT, AS C1 HAS TO GO ON TOP AND BE BENT OVER.

CAPACITORS: ALL CAPACITORS MUST BE BENT 90 DEG'S TOWARDS THE SIDES JUST CLEARING EDGE BEFORE SOLDERING IN TO REDUCE HEIGHT.

C1-C3 & C5: FIT THE LEADS IN FROM THE TOP AND BEND OVER TOWARDS THE EDGE AND SOLDER IN.

C4: FIT A BIT OF SLEEVING OVER THE NEGATIVE LEG TO AVOID THE POSSIBILITY OF SHORTING AND BEND THE LEG AS DEPICTED IN THE DIAGRAM AND SOLDER IT IN PLACE.

MAX 232: SOLDER IN MAX232 CHIP WITHOUT A SOCKET AGAIN TO REDUCE HEIGHT. PUT VERO BOARD MODULE NEAR 44 WAY EDGE CONNECTOR TO GET IDEA FOR THE LENGTH OF THE 4 CONNECTING WIRES WHICH YOU SOLDER IN NEXT. AT THIS TIME YOU CAN ALSO SOLDER ON THE 3.5 MM STEREO SOCKET.

74LS74: BEND ALL PINS EXCEPT PIN 7 & 14 AT RIGHT ANGLES AND PLACE IT OVER U3-74LS32 IN DISK CONTROLLER AND SOLDER PINS 7 & 14 OF THE 74LS74 TO IT WHICH SUPPLY GND AND +5V TO THE 74LS74. MAKE SURE PIN 7 OF 74LS74 IS SOLDERED TO PIN 7 OF 74LS32 AND THAT PIN 14 IS SOLDERED TO PIN 14.

WITH A SHORT LENGTH OF BARE WIRE JOIN PINS 1, 2, 3, 4 & 13 TO PIN 14 AS SHOWN ON THE DIAGRAM. SOLDER THE 4K7 RESISTOR ON NEXT AND TRIM THE LEADS. CONNECT THE REMAINING WIRES BETWEEN U4, U6, U7, PIGGY BACKED 74LS74 AND MAX232 MODULE NEXT. IT SHOULD BE EASIER FOLLOWING WIRING DETAILS ON THE DIAGRAM THAN A BLOW BY BLOW DESCRIPTION.

CHECK ALL YOUR WORK FOR SHORTS OR ERRORS ESPECIALLY BETWEEN IC PINS. WRAP MAX232 MODULE IN THIN CARDBOARD TO PREVENT SHORTS AND FIT 3.5 MM STEREO SOCKET IN THE BOTTOM RIGHT ABOUT 3 CM'S FROM THE REAR. AFTER TESTING UNIT OUT AND ALL OK AND IF FITTING RF SHIELD BACK ON THEN THE FRONT OF THE SHIELD WILL HAVE TO BE BENT UP 90 DEG'S TO CLEAR MAX232 MODULE.

TESTING: THIS IS GOING TO BE THE DIFFICULT PART IF YOU HAVEN'T A SECOND VZ WITH DRIVE AND DISK CONTROLLER. IT WILL WORK WITH ANOTHER COMPUTER PROVIDING IT HAS A RS-232 PORT. THE COMMODORE 64 IS NOT SUITABLE AS THE SIGNALS ARE INVERTED. I HAVEN'T BUILT MY UNIT YET, SO I CAN'T ADVISE ON TESTING EXCEPT THAT YOU FOLLOW INSTRUCTIONS SENT WITH THE SOFTWARE.

U7-74LS125: IT HAS FOUR TRISTATE BUFFERS AND ONLY THREE ARE USED BY THE DISK CONTROLLER. THE BLOCK DIAGRAM DEPICTS ALL BUFFERS INCLUDING THE SPARE ONE WHICH WE CAN PRESS INTO SERVICE.

Piggy-backing: THE DIAGRAM SHOWS HOW THE PINS ARE BENT OUT AND CHIP SOLDERED ONTO THE ONE BELOW. IN PRACTICE THE CHIPS WILL BE CLOSER THAN DEPICTED.

U2-74LS138: THE DISK CONTROLLER ONLY USES FOUR OUTPUTS, Q0-Q3 WITH Q4-Q7 BEING SPARE AND PETER AND ANDY HAVE CLEVERLY UTILISED TWO OF THE UNUSED OUTPUTS.

74LS74: FOR A SINGLE RS-232 PORT ONLY THE MAX232 AND 74LS74 CHIPS ARE NEEDED WITH THE UNUSED INPUTS ON THE LATTER TAKEN TO +5V TO PREVENT POSSIBLE OSCILLATION.

3.5 mm STEREO SOCKET: A ROUND METAL TYPE AS SOLD BY D. SMITH IS OK AND IS THE ONE WHICH I'VE CHOSEN FOR MY OWN USE. FILE THE FLANGE DOWN ON ONE SIDE SO IT CAN SIT FLUSH ON THE BOTTOM OF THE CASE. TRY TO GET ONE WITH A NUT AS THEY ARE EASIER TO FIT.

CASES: IF BUILDING YOUR UNIT INTO A CASE THEN I WOULDN'T ADVISE THE ONES FROM D. SMITH AS THEY HAVE INTERNAL RIBBING WHICH MAKE IT DIFFICULT TO MOUNT SOCKETS, SWITCHES, ETC. THE ONES FROM TANDY DON'T SUFFER FROM ABOVE PROBLEM.

USING YOUR RS-232 PORT: PETER AND ANDY HICKMAN HAVE PROVIDED VZ USERS WITH A RS-232 PORT TOGETHER WITH A PROGRAM TO DRIVE IT. AT THIS STAGE I KNOW OF NO OTHER PROGRAM AROUND FOR IT. THE PORT CAN'T BE USED WITH A PRINTER WITH A SERIAL INPUT PORT SIMPLY BECAUSE AT THIS TIME THERE'S NO PRINTER PATCH OR HARDWARE INTERFACE THAT I KNOW OF FOR IT.

IN MY OWN CASE I'LL BE USING IT TO TRANSFER DATA BACK AND FORTH BETWEEN THE VZ AND IBM PC COMPATIBLE WHICH HAS A SPARE RS-232 PORT AND WITH A MODEM WHICH I HOPE TO ACQUIRE BEFORE CHRISTMAS. AT OUR LAST CLUB MEETING PETER JONES DEMONSTRATED HIS DUAL VERSION WHICH WILL ALSO BE SHOWN AT THE NEWCASTLE COMPUTER SHOW.